M4010104 / M4020104: PROGRAMMING IN PYTHON · DIGITAL UNIVERSITY KERALA · CAPSTONE PROJECT SPECIFICATION

Project 20: Cardiovascular Heart Failure Biomarker Profiling & Mortality Risk Scorer

Effort Budget: 10–12 Hours | Core Stack: Pure Python, NumPy, Pandas, Matplotlib, Seaborn, Plotly | Max Marks: 25 | Instructor: Dr. Sunil TT

1. Executive Overview & Domain Context

Cardiovascular diseases represent the leading cause of global mortality. Clinical diagnostic records compile physiological metrics—including resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiogram (ECG), maximum heart rate achieved (MaxHR), exercise-induced angina, and ST-segment depression (Oldpeak)—to evaluate cardiac ischemia and heart failure risk.

Your team will architect an interpretable cardiac risk profiling package in pure Python that cleans patient physiological logs, handles zero-cholesterol measurement dropouts via median imputation, computes standardized effect sizes (Cohen's d) across biomarkers in NumPy, models multi-factor cardiac risk scores, and visualizes patient risk answering the 12 core questions below.

Primary Public Dataset & Telemetry/Data Schema

The project utilizes the verified public Heart Failure Clinical Records Dataset on Kaggle (**`https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction`**). The dataset structure captures:

- **Raw Feature Attributes:** **`Age`**, **`Sex`**, **`ChestPainType`** (TA, ATA, NAP, ASY), **`RestingBP`**, **`Cholesterol`**, **`FastingBS`**, **`RestingECG`**, **`MaxHR`**, **`ExerciseAngina`**, **`Oldpeak`**, **`ST_Slope`**, **`HeartDisease`**.
- **Engineered / Target Variables:** **`cohens_d_scores`**, **`composite_cardio_risk`**, **`cholesterol_imputed`**, **`risk_percentile_tier`**.

2. Complete Project Directory & File Architecture

Each team must organize their repository into a standardized, production-grade layout. All application source code must reside inside a dedicated package (**`src/`**), with separate directories for raw/curated data, interactive development notebooks, visual figures, and formal documentation:

```
cardiovascular-risk-profiler/
├── README.md                          # Repository documentation & GitHub Pages source
├── requirements.txt                   # Dependency specification (numpy, pandas, matplotlib, seaborn, p
├── run_pipeline.py                    # Standalone production CLI application entry point
├── data/
│   ├── raw/
│   │   └── heart_failure_clinical_raw.csv            # Raw uncleaned input dataset (including simulated
│   └── processed/
│       ├── clean_heart_failure.csv          # Curated, validated clean CSV dataset
│       └── cardio_clinical_audit.log           # Timestamped audit report detailing filtered/rejected r
├── src/
│   ├── __init__.py                    # Python package initialization
│   ├── ingestion.py                   # Defensive line-by-line file reader, range auditor & CSV exporte
│   ├── analyzer.py                    # CardioRiskProfiler OOP class, NumPy vectorization & Pandas anal
│   └── visualizer.py                  # Matplotlib envelope plots, Seaborn heatmaps & Plotly HTML expor
├── notebooks/
│   └── development_cycle.ipynb        # Development cycle notebook: EDA, trials, tests & unit assertion
└── reports/
    ├── figures/
    │   ├── primary_envelope.png       # Matplotlib confidence envelope & regression overlay
    │   ├── correlation_heatmap.png    # Seaborn annotated feature correlation heatmap
    │   └── distribution_regimes.png   # Seaborn distribution across categorical/domain tiers
```

```
    ├── interactive_dashboard.html     # Standalone interactive Plotly HTML dashboard with controls
    └── Project_Report.pdf             # 3-4 page final technical report
```

3. Dual Development Deliverables: Notebook vs. Standalone Program

To mirror industry software engineering standards, each student group must deliver two distinct, complementary implementations of their solution:

Deliverable A: The Development Cycle Notebook (`notebooks/development_cycle.ipynb`)

The development notebook chronicles your team's engineering journey from raw data exploration to validated model prototypes. It is not merely a final script pasted into cells, but an authentic record of the development cycle:

- **Phase 1: Exploratory Data Ingestion & Diagnostics:** Loading raw data, inspecting missing values, testing boundary rules, and logging rejected records.
- **Phase 2: Mathematical Formulation & Feature Verification:** Developing and validating NumPy array calculations against target benchmarks using `np.allclose()`.
- **Phase 3: OOP Class Architecture & Unit Tests:** Instantiating the `CardioRiskProfiler` class, executing automated unit assertions (e.g., verifying that passing invalid parameters raises an explicit `ValueError`), and testing edge cases.
- **Phase 4: Statistical Modeling & Regimen Exploration:** Implementing ordinary least-squares regression, rolling-window calculations, and multi-metric Pandas grouping.
- **Phase 5: Visual Prototype Development:** Iteratively designing and refining Matplotlib and Seaborn figures, followed by configuring the interactive Plotly HTML dashboard.

Deliverable B: The Final Standalone Python Program (`run_pipeline.py` & `src/`)

While the notebook documents the exploratory development cycle, your repository must also provide a production-ready, fully executable Python application. The program must run end-to-end directly from the terminal without requiring Jupyter:

- **Command-Line Interface (CLI):** Executed via `python run_pipeline.py --input data/raw/heart_failure_clinical_raw.csv --output-dir data/processed --report-dir reports`.
- **Modular Package Import:** The driver script imports cleanly separated domain logic from `src.ingestion`, `src.analyzer`, and `src.visualizer`.
- **Terminal Feedback & Exit Codes:** The program must print structured progress indicators, log clean/faulty record counts to the console, and return an exit code of `0` upon successful pipeline completion.
- **Automated Artifact Generation:** In a single execution pass, the script reads raw inputs, writes `data/processed/clean_heart_failure.csv`, saves `data/processed/cardio_clinical_audit.log`, exports static figures to `reports/figures/`, and generates the standalone `reports/interactive_dashboard.html`.

4. Technical Investigation Requirements & Research Questions (Q1–Q12)

Your team's implementation across the development notebook and standalone program must rigorously answer the following 12 course-aligned engineering questions:

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
File I/O & Exceptions (`src/ingestion.py`)	**Q1**	**Cardiovascular Vitals & Zero-Measurement Audit:** Parse raw patient CSV lines with try-except. Detect invalid resting blood pressure (< 50 or > 250 mmHg), handle zero-cholesterol records, validate age bounds, and log errors.	open(), try-except (ValueError, FileNotFoundError), physiological checks

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Data Curation & Export (`src/ingestion.py`)	**Q2**	**Curated Cardiac Clinical Export & Audit Log:** Write clean patient records to clean_heart_failure.csv while logging anomalous zero-cholesterol lines and malformed categorical fields in cardio_clinical_audit.log.	Defensive file writing, dictionary parsing, formatted log reporting
OOP Design (`src/ingestion.py`)	**Q3**	**Cardio Risk Profiler Class:** Design an encapsulated class CardioRiskProfiler with methods load_from_csv(), compute_cohens_d(biomarker='Oldpeak'), and summary_report(). Ensure invalid biomarker raises ValueError.	OOP class, __init__, encapsulation, raise ValueError
NumPy Arrays (`src/analyzer.py`)	**Q4**	**Vectorized Cohen's d Effect Size Verification:** Using only raw NumPy arrays of heart-disease positive vs. negative cohorts, compute Cohen's d effect sizes for Oldpeak, MaxHR, and Age. Verify values via np.allclose().	np.mean(), np.var(), pooled std, np.allclose()
NumPy Masking (`src/analyzer.py`)	**Q5**	**High-Risk Cardiac Anomaly Masking:** Flag patients exhibiting both high ST depression (Oldpeak > 2.0) and Exercise Angina in NumPy. Determine overlap with asymptomatic chest pain cohorts (ASY) via boolean masks.	np.mean(), np.std(), boolean masks (&, |), Python set
NumPy Regression (`src/analyzer.py`)	**Q6**	**Oldpeak and MaxHR vs. Heart Disease OLS Model:** Fit an ordinary least-squares model (HeartDisease = b0 + b1*Oldpeak + b2*MaxHR + b3*Age) via np.linalg.lstsq. Quantify relative predictive weights of ST depression vs. heart rate.	np.column_stack(), np.linalg.lstsq(), matrix product (@), R2
Pandas Series (`src/analyzer.py`)	**Q7**	**Chest Pain Type & Cholesterol Missingness Audit:** Quantify patient counts across ChestPainType categories in Pandas. Evaluate the impact of median cholesterol imputation on disease correlation.	.groupby('ChestPainType').agg(), .value_counts()
Pandas Groupby (`src/analyzer.py`)	**Q8**	**Age-MaxHR Discretization & Prevalence Pivot Table:** Discretize Age into 40-50, 50-60, 60+; discretize MaxHR into Low (< 120), Normal (120-150), High (> 150). Build a 3×3 pivot_table(margins=True) showing heart disease rates.	pd.cut(), .groupby().agg(), .pivot_table(margins=True)
Pandas Reshaping (`src/analyzer.py`)	**Q9**	**Wide-to-Long Cardiac Biomarker Ranking via Melt:** Reshape continuous biomarkers (RestingBP, Cholesterol, MaxHR, Oldpeak) into long format via pd.melt() and extract top 5 most severe risk profiles.	pd.melt(), .groupby('Biomarker'), .nlargest(5)
Matplotlib (`src/visualizer.py`)	**Q10**	**Matplotlib MaxHR vs. Oldpeak Clinical Scatter Plot:** Plot MaxHR vs. Oldpeak in Matplotlib with disease status colored (red = disease, green = normal). Overlay diagnostic threshold lines and custom legends.	plt.subplots(), ax.scatter(), ax.axvline(), ax.axhline()
Seaborn (`src/visualizer.py`)	**Q11**	**Seaborn Clinical Correlation Heatmap & ST Slope Boxplot:** Render an annotated correlation heatmap across numerical vitals with sns.heatmap(). Create a side-by-side boxplot comparing Oldpeak distributions across ST_Slope types.	df.corr(), sns.heatmap(), sns.boxplot(), sns.despine()

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Plotly HTML (`src/visualizer.py`)	**Q12**	**Interactive Cardiac Risk Triage Dashboard:** Build an interactive HTML dashboard (interactive_dashboard.html) with 2D/3D clinical scatter plots, chest pain dropdown filters, and patient risk percentile sliders.	plotly.express.scatter(), custom hover tooltips, write_html()

5. Collaborative Git Workflow, GitHub Pages & README.md Specification

Best Practices: Collaborative Git Workflow & GitHub Pages Deployment from Day 1

From the very first day, the student team must adopt professional version control and documentation practices. On Day 1, one designated team member initializes the central public GitHub repository, establishes a clear branch protection rule preventing direct pushes to `main`, adds all teammates as registered **Contributors** (with write access), and grants read access to the Course Instructor (`Dr. Sunil TT`) and teaching assistants. Immediately thereafter, the team must activate **GitHub Pages** (via *Settings → Pages → Build and deployment: Deploy from a branch: main /docs or root*) so that project documentation, interactive architecture diagrams, and the live Plotly dashboard (`reports/interactive_dashboard.html`) are publicly accessible and continuously deployed throughout the development lifecycle. All feature development must follow a strict branching model (e.g., `feature/ingestion`, `feature/analysis`, `feature/visualization`): individual team members must write small, atomic, well-commented commits from their own personal GitHub accounts and merge them into `main` exclusively through formal **Pull Requests (PRs)** accompanied by peer review. Dumping entire codebases in a single, monolithic end-of-semester commit is strictly prohibited and will result in zero marks for individual Git contribution; maintaining an unbroken, transparent Git commit and PR audit trail is essential for passing the code ownership defense during the live viva-voce examination.

Repository README.md Specification Template

The root `README.md` serves as the primary technical landing page and must strictly conform to the following structural format:

````
# Project 20: Cardiovascular Heart Failure Biomarker Profiling & Mortality Risk Scorer
[![Course](https://img.shields.io/badge/Course-Programming%20in%20Python-blue.svg)](https://duk.ac.in)
[![Python](https://img.shields.io/badge/Python-3.11%2B-brightgreen.svg)](https://python.org)
[![License](https://img.shields.io/badge/License-MIT-yellow.svg)](LICENSE)

## 1. Project Overview & Domain Context
- **Institution:** Digital University Kerala (MTech AI / MSc AI)
- **Course:** M4010104 / M4020104 — Programming in Python | **Instructor:** Dr. Sunil TT
- **Dataset:** Heart Failure Clinical Records Dataset on Kaggle (https://www.kaggle.com/datasets/fedesor

## 2. System Architecture & Pipeline Design
[Provide an ASCII or Mermaid architecture flow diagram showing raw ingestion -> validation -> analyzer -

## 3. Installation & Environment Setup
```bash
git clone https://github.com/<your-team>/cardiovascular-risk-profiler.git
cd cardiovascular-risk-profiler
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

## 4. Execution Instructions
### A. Running the Standalone Application
```bash
python run_pipeline.py --input data/raw/heart_failure_clinical_raw.csv --output-dir data/processed --rep
```
### B. Running the Development Notebook
```bash
````
````

```
jupyter notebook notebooks/development_cycle.ipynb
```

5. Key Analytical & Statistical Findings (Q1-Q12 Summary)
[Provide a concise markdown summary table detailing raw vs. clean counts, regression coefficients, anoma

6. Interactive Telemetry Dashboard
Explore the live interactive Plotly visualization hosted on GitHub Pages:
🔗 [Live Project Dashboard](https://<your-team>.github.io/cardiovascular-risk-profiler/interactive_dashb

7. Contributor Git Commit & Pull Request Log Table
Contributor Name	GitHub Handle	Assigned Module / Tasks	Verified Commit Hash Range	Merged Pull
[Student Full Name]	`@github_handle`	Modular Architecture (`src/ingestion.py`, `src/analyzer.py`,

6. Evaluation Rubric & Mandatory Academic Integrity Governance

Projects are evaluated on a 25-mark scale based on code quality, mathematical correctness, visualization depth, git collaboration discipline, and live technical defense:

Assessment Criterion	Basis of Evaluation & Deliverable Verification	Max Marks
1. Data Curation & File I/O Pipeline (Q1–Q2)	Robust line-by-line file reading with defensive **`try-except`** handling, physical/domain bounds validation, clean CSV output, and structured **`cardio_clinical_audit.log`** reporting.	**5 Marks**
2. OOP, NumPy & Pandas Analytics (Q3–Q9)	Modular **`CardioRiskProfiler`** class architecture, vectorized NumPy feature re-derivations (verified via **`np.allclose`**), regression modeling, and Pandas rolling/groupby/melt operations.	**6 Marks**
3. Multi-Library Visualizations (Q10–Q12)	Publication-quality Matplotlib confidence envelope/trend plot, Seaborn correlation heatmap and distribution violin plot, and standalone interactive Plotly HTML dashboard.	**5 Marks**
4. Software Packaging, GitHub & Documentation	Clean directory layout, comprehensive **`README.md`**, working GitHub Pages site, **`development_cycle.ipynb`** notebook, standalone **`run_pipeline.py`**, and balanced individual Git commits/PRs.	**4 Marks**
5. Live Presentation & Code Viva-Voce Defense	Technical walkthrough and individual live viva defense where each student must explain or modify lines of code in real time without external assistance.	**5 Marks**

> **STRICT ACADEMIC INTEGRITY & ZERO-TOLERANCE AI POLICY (0 MARKS PENALTY):** All submitted Python source code (**`.py`**, **`.ipynb`**) and written documentation will be screened through an automated **AI-Code & Plagiarism Detection Suite**. If AI-generated code or synthetic documentation is detected at any checkpoint or final submission, **0 Marks will be awarded immediately** for the entire project. Each student must prove authentic individual code ownership during the live viva-voce examination — *Course Instructor: Dr. Sunil TT*.

7. Milestone Schedule, Submission Checklist & Viva Defense Prompts

To ensure systematic progress across the semester, teams must adhere to three project checkpoints and prepare for on-the-spot live code modification challenges during the viva-voce examination:

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 1 (Data Ingestion)	Week 11 Lab Session	Repository initialized with contributors and GitHub Pages active. **`src/ingestion.py`** operational with line-by-line filtering, **`clean_heart_failure.csv`** exported, and **`cardio_clinical_audit.log`** generated.

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 2 (Analytics & OOP)	Week 13 Lab Session	**`src/analyzer.py`** complete with **`CardioRiskProfiler`** class, NumPy feature checks (Q4), outlier masks (Q5), regression modeling (Q6), and exploratory notebook sections.
Final Submission & Viva	Week 15 Project Review	Full package operational: standalone **`run_pipeline.py`**, polished **`development_cycle.ipynb`**, interactive **`interactive_dashboard.html`** on GitHub Pages, 3–4 page PDF report, and live code defense.

Representative Live Viva-Voce Code Modification Challenge

During the individual code viva-voce, the instructor will ask students to execute an immediate 2-minute code modification on their working repository to demonstrate genuine authorship:

- **Challenge Prompt:** *"In src/analyzer.py, calculate Cohen's d effect size for Oldpeak vs. MaxHR. Re-run your module and state which biomarker demonstrates stronger diagnostic separation between positive and normal cohorts."*